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Casing driver and method

Abstract

An apparatus and method are provided for using a rotary drive of a drilling rig to drive both the drill stem and the casing simultaneously, by connecting the drill stem to the casing using a drive head. Subsequently, after drilling a first length of bore, the drive head may be removed and the drill stem driven by the rotary drive independently from the casing, allowing the bore to be extended further without moving the casing.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Background/Summary

BACKGROUND

(1) Typically, when using a drilling rig for partially cased auger cast or auger cast displacement piles, the drilling rig rotary drive is connected directly to both the drill stem and the casing. Casing is used as required over the drill tool as the drilled hole is advanced through a first layer and not needed for subsequent layers. This typical connection of the casing to the rotary drive limits pile depth beyond the tip of the casing to the length of the drill stem that extends above the rotary.

(2) Thus, what is needed is an apparatus and method which accommodate the use of casings having different lengths on a full length drill stem by connecting the casing drive to the drill string, below the rotary, which enables the drilling rig to achieve its maximum drilling depth with shorter lengths of casing. The limitation of the typical connection is thereby removed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments are illustrated by way of example and not limitation in the accompanying drawings, in which like references indicate similar elements, and in which:

(2) FIG. 1 is a side view of a drilling rig employing embodiments of a casing driver;

(3) FIG. 2 is a side view of a drilling rig employing embodiments of a casing driver;

(4) FIG. 3 is a side view of a drilling rig employing embodiments of a casing driver;

(5) FIG. 4 is a side view of a drilling rig employing embodiments of a casing driver;

(6) FIG. 5 is a side view of a drilling rig employing embodiments of a casing driver;

(7) FIG. 6 is a side view of a drilling rig employing embodiments of a casing driver;

(8) FIG. 7 is a side view of a drilling rig employing embodiments of a casing driver;

(9) FIG. 8 is a side view of a first embodiment of a casing driver;

(10) FIG. 9A is a side view of aspects of the first embodiment of a casing driver;

(11) FIG. 9B is a top view of the section indicated in FIG. 8A;

(12) FIG. 10A is a bottom isometric view of the first embodiment of a casing driver when inverted;

(13) FIG. 10B is an upper isometric view of the first embodiment of a casing driver;

(14) FIG. 11 is an upper isometric view of the first embodiment of a casing driver disassembled;

(15) FIG. 12 is a side view of the first embodiment of a casing driver;

- (16) FIG. 13A is an upper isometric view of aspects of embodiments of a casing driver;
- (17) FIG. 13B is a lower isometric view of aspects of embodiments of a casing driver;
- (18) FIG. 14 is an upper isometric view of a second embodiment of a casing driver;
- (19) FIG. 15A is a side view of aspects of the second embodiment of a casing driver;
- (20) FIG. 15B is a top view of the section of the embodiment of a casing driver indicated in FIG. 15A;
- (21) FIG. 15C is a side view of aspects of the second embodiment of a casing driver;
- (22) FIG. 16 is an upper isometric view of aspects of the second embodiment of a casing driver;
- (23) FIG. 17 is an upper isometric view of the second embodiment of a casing driver disassembled;
- (24) FIG. 18 is an upper isometric view of aspects of the second embodiment of a casing driver;
- (25) FIG. 19 is a side view of the second embodiment of a casing driver;
- (26) FIG. 20 is an upper isometric view of aspects of the second embodiment of a casing driver disassembled; and
- (27) FIG. 21 is a side view of the second embodiment of a casing driver.

DETAILED DESCRIPTION

(28) Embodiments described within disclose an apparatus and method using an embodiment of a casing driver connected to different points on the drill stem. Thus, embodiments facilitate using a variety of casing lengths for a given drill stem length. Furthermore, in embodiments, the rotary drive may be connected only to the drill stem, with both drill stem and casing being driven by that single connection to the rotary. These embodiments will make it possible to construct drilled foundation piles, including auger cast piles and auger cast displacement piles, in soil conditions that require the use of casing in an upper section of the drilled hole, but where casing is not required in the lower section of the drilled hole. Specifically, the embodiments will enable a drill operator to assemble a drill stem of sufficient length to drill a collection of holes with varying depths and soil conditions and interchange different casing lengths according to the drill depths and soil conditions at each respective hole in the collection. The interchange and use of varying lengths of casing for different holes will not require disassembling the drill stem.

(29) FIG. 1 through FIG. 6 are side views of a drilling rig 10 employing embodiments of a casing driver 100, e.g., a casing driver as described with regard to casing driver 101 (FIG. 8-FIG. 12) and casing driver 102 (FIG. 14-FIG. 21). FIG. 1 through FIG. 6 illustrate methods of using embodiments of casing driver 100. In the embodiment shown in FIG. 1 through FIG. 6, rotary drive 50 has a center through hole that allows the drill stem 40 to pass through the rotary drive 50. Rotary drive 50 may engage the drill stem 40 at different points along drill stem 40. Drill stem 40 passes through a casing 60 and terminates with a drilling tool 70, e.g., a displacement tool with an auger 72. Rotary drive 50 may engage the drill stem 40 at designated locations along drill stem 40 and be used to cause drill stem 40 to rotate about its central axis, causing a drilling tool 70 to rotate as well and create a bore 80. Once bore 80 is completed, and as drilling tool 70 is being removed, a grout, such as concrete, may be supplied to grout conduit 42, and through drill stem 40 into bore 80. A stroke 90 indicates a distance that rotary drive 50 may be lowered and raised and thus represents a bore depth that may be attained when rotary drive 50 travels a single stroke 90. To create a bore with a depth greater than stroke 90, rotary drive 50 may be used to drive drilling tool 70 a first stroke length into the ground. Then, rotary drive 50 may be moved higher up on drill stem 40 and used to force drilling tool 70 further into the soil. Thus, bore 80 may be drilled to a desired depth. In addition, multiple drill stems 40 may be joined together to increase the depth of bore 80 past that of a single drill stem length.

(30) In embodiments, casing driver 100 may be connected to drill stem 40, so that as rotary drive 50 forces drill stem 40 downward or upward or rotates, the downward, upward, or rotational force is transmitted from drill stem 40, through casing driver 100 and into a cone 62 and casing 60. Thus, as drill stem 40 is forced downward or upward or rotated, casing 60 travels or rotates with it. Casing 60 may be disengaged from drill stem 40 at an appropriate depth by disengaging casing

driver **100** from drill stem **40**. In embodiments, casing driver **100** abuts cone **62** and transfers axial and torsional forces from drill stem **40** to casing **60**.

(31) In FIG. **1**, rig **10** has just begun drilling bore **80**. Casing **60** has been assembled around the drill stem **40** and casing driver **100** is engaged to drill stem **40** at a connection point on drill stem **40**, which is configured to engage casing driver **100**. Casing driver **100** is configured to engage and transmit torsional, upward, and downward forces into cone **62** and into casing **60**. Accordingly, with casing driver **100** engaged, casing **60** and drill stem **40** are ready to travel upwards and downwards or rotate together in unison. FIG. **2** illustrates that, as rig **10** drives rotary **50** downward with casing driver **100** engaged to drill stem **40**, drill stem **40**, casing driver **100**, casing **60**, and drilling tool **70** travel downward and rotate in unison. Further, FIG. **2** shows that casing **60** has reached a desired casing depth. and, at this point casing driver **100** can be removed so that the remainder of the bore **80** can be drilled without casing **60**. In other words, after removing casing driver **100**, drill stem **40** may travel independently of casing **60** such that drilling can continue without advancing the casing **60** further into bore **80**.

(32) FIG. **3** shows that the drill stem and drilling tool **70** have advanced downward to complete bore **80**, which would occur after casing driver **100** has been removed or disconnected. Once reaching the bore **80** bottom tip elevation, the completed excavation includes upper cased section **66** and lower uncased section **68**. In FIG. **3**, casing driver **100** has been removed from drill stem **40** and casing **60**. Rotary drive **50** has been relocated to the upper end of drill stem **40**. Subsequently, drilling rig **10** has used rotary drive **50** to advance drill stem **40** through casing **60** such that drilling tool **70** has drilled bore **80** a significant depth below casing **60** and has reached the desired depth for bore **80**. The completed bore **80** now comprises an upper and lower section. The upper section **66** of bore **80** may be referred to as a “cased section” and the drilling operation may be described as cased drilling or drilling with casing. The bottom section **68** of bore **80**, which is drilled without casing, may be referred to as an “uncased section” of bore **80** and the drilling operation may be described as uncased drilling or drilling without casing. This illustrates a feature provided by embodiments in which the length of a casing may be different from the length of a drill stem, and the desired depths for the uncased section **68** of bore **80** may exceed the maximum allowable length of drill stem **40** extending above a rotary drive **50** for a given drill rig **10** and this is accomplished without adding sections of drill stem to drill stem **40**.

(33) In FIG. **3**, casing driver **100** has been completely removed. However, in embodiments, casing driver **100** may simply be disconnected from drill stem **40** (and remain atop cone **62**), such that drill stem **40** may move with respect to both casing driver **100** and casing **60** without imparting that movement to casing **60**. Once casing driver **100** has been removed, the drill stem **40** and drilling tool **70** may advance independently of casing **60**.

(34) In FIG. **4**, drilling rig **10** is withdrawing drill stem **40** and drilling tool **70** from bore **80**. As drilling tool **70** is being withdrawn, grout **82** is pumped through drilling rig **10** and into bore **80**. Grout **82** is pumped through grout conduit **42** (external tubing associated with supplying grout to tubing **42** is not shown for simplicity). The grout travels through a hollow center pipe in drill stem **40** and drilling tool **70** and into bore **80**. Drill stem **40** and drilling tool **70** continue traveling upward while pumping grout **82** into bore **80** until a desired connection point on drill stem **40** is just above cone **62**, at which point casing driver **100** is re-engaged around drill stem **40** at the connection point on drill stem **40** such that casing **60** and drill stem **40** are ready to travel upwards together in unison. Pumping grout may pause briefly to give support personnel time to re-engage the casing driver **100** and pumping resumes just before casing **60** and drill stem **40** begin advancing upwards together.

(35) In some embodiments, drill stem **40** will be provided with a plurality of connection points configured to engage casing driver **100** and spaced longitudinally along the long axis of drill stem **40**. There may also be provided many different sections of casing **60** each having cone **62** fixed to one end and each individual section of casing **60** having a different length measured along its long

axis. In these embodiments, when soil conditions or structural designs require varying lengths of upper cased sections between holes, a drill operator may interchange the different sections of casing **60** to match the casing length requirements of each individual hole to the length of each individual section of casing **60**.

(36) In FIG. 5, drill string **40** has been withdrawn fully back into casing **60** such that drilling tool **70** is positioned at a lower distal end of casing **60** and a desired connection point on drill stem **40** is positioned just above cone **62** to allow casing driver **100** to be re-installed. In FIG. 5, casing driver **100** has been re-installed back on drill string **40** and engages cone **62**, allowing the further upward travel of drill string **40** to withdraw both drill stem **40** and casing **60** as one, with grout **82** being supplied to fill bore **80** as drill stem **40** and casing **60** are raised.

(37) In FIG. 6, drill stem **40** and casing **60** have been completely removed from bore **80**, which is shown completely filled with grout. Bore **80** is complete and grouted when casing **60**, drill stem **40**, and drilling tool **70** are removed from bore **80**.

(38) FIG. 7 is a side view of drilling rig **10** employing embodiments of a casing driver **100** that include a perforated spacer **130**. Perforated spacer **130** may be positioned between casing driver **100** and casing **60** (as shown in FIG. 7, FIG. 12, and FIG. 21) and the arrangement used as described with respect to FIG. 1 and FIG. 6. Perforated spacer **130** allows the spoils of the drilling process to pass from within casing **60** and exit through the perforations. Thus, embodiments of casing driver **100** with perforated spacer **130** may be used where spoils travel between the drill stem and casing.

(39) Thus, a method of using an embodiment of a casing driver connected to the drill string may include: a step of connecting a rotary drive of a drilling rig to a drill stem at a first longitudinal location on the drill stem, the drill stem including a drill bit at a distal end; a step of connecting a drive head to the drill stem at a second longitudinal location on the drill stem separated from the first longitudinal location; a step of placing a casing about the drill stem with a drive head, the casing extending from the drive head connected to the drill stem toward the drill bit; and a step of drilling a first length of a bore with the drill stem and casing moving into the bore and without relative axial motion between the casing and pipe. The method may further comprise, after drilling the first length of the bore: a step of disconnecting the drive head from the drill stem; and a step of drilling a second length of the bore with the drill stem moving axially within the casing and the casing remaining within the first length of the bore. The method may further comprise a step of pumping grout into the bore; while pumping grout, a step of lifting the drill stem out of the second length of the bore with the drill stem moving axially within the casing at a rate that matches the rate grout is filling the bore; a step of reconnecting the drive head to the drill stem; and while pumping grout into the bore, a step of lifting the drill stem, casing out of the first length of the bore and without relative axial motion between the casing and drill stem.

(40) FIG. 8 is a side view of a first embodiment **101** of casing driver **100**. Casing driver **101** is adapted to connect to a drill stem **40a** that is round in cross-section (FIG. 8B). In FIG. 8, casing driver **101** includes a drill stem attachment **110** and a casing interface **120**. Bar attachment **110** is connected to drill stem **40a** so that both rotation and axial movement of drill stem **40a** is imparted to drill stem attachment **110**. Drill stem **40a** is shown to pass into and through casing **60**.

(41) In the embodiment, drill stem attachment **110** is fixed to casing interface **120**. As shown in FIG. 11, drill stem attachment **110** and casing interface **120** each comprise identical interface halves configured to wrap around and bolt to drill stem **40a** as shown in FIG. 8. Thus, linear and torsional forces are transmitted from stem **40a** through drill stem attachment **110** to casing **60**, causing casing **60** to travel with drill stem **40a** into the bore.

(42) Casing interface **120** includes posts **122** that extend radially and are configured to be received by slots **64**, e.g., T-slots, within cone **62**. Thus, in the embodiment, casing interface **120** partially enters cone **62** to the depth that posts **122** descend into slots **64** and linear and torsional forces may be transmitted from casing interface **120** into cone **62** and casing **60**. The connection between cone

62 and casing 60 may include welding, machine threading, and/or bolting.

(43) FIG. 9A is a side view of aspects of drill stem 40a, configured to be used with casing driver 101. FIG. 9B is a top view of the section indicated in FIG. 9A. In FIG. 9A and FIG. 9B, drill stem 40a is shown to be provided with plates 44 that are configured to receive fasteners that connect drill stem attachment 110 to drill stem 40a. FIG. 9A illustrates a single section of drill stem 40a with an upper connection and a lower connection. Multiple such sections may be combined, one atop the other, depending on the desired depth of bore 80. In addition, while FIG. 9A illustrates two sets of plates 44 about drill stem 40a, in embodiments, sets of plates 44 may be distributed along drill stem 40a at arbitrary locations according to the desires of the operator. Providing two or more plates 44 spaced along the long axis of a drill stem 40a enables the use of varying lengths of casing 60 without a drill operator needing to reconfigure the assembled drill stem.

(44) FIG. 10A is an isometric view of casing driver 101 when inverted. FIG. 10B is an upper isometric view of casing driver 101. In FIG. 10A, casing interface 120 is shown to include identical interface halves 121a and 121b. Each half 121a, 121b includes an outer ring segment 124 and an inner ring segment 126. Inner ring segment 126 is connected to outer ring segment 124 by posts 122, which are inserted into a window of inner ring segment 126 and received by a slot in outer ring segment 124. Halves 121a, 121b are joined together using fasteners through flanges 128a, 128b.

(45) Drill stem attachment 110 includes identical attachment halves 112a, 112b. Each half 112a, 112b includes an upper flange 116a and a lower flange 116b connected by connector plates 114a, 114b and by connector flanges 118a, 118b. Halves 112a, 112b are joined together, e.g., using fasteners through holes 119, in flanges 118a, 118b. The identical attachment halves 112a, 112b of drill stem attachment 110 are each welded to the respective identical interface halves 121a and 121b of casing interface 120.

(46) In FIG. 10A, posts 122 are shown to taper from inner ring section 126 to outer ring section 124. This taper helps center casing interface 120 and drill stem 40a within cone 62 when casing driver 101 is being lowered toward casing 60.

(47) FIG. 11 is an upper isometric view of casing driver 101 partially disassembled with halves 112a, 112b separated. In this configuration, halves 112a, 112b may be placed about drill stem 40a and fasteners passed through connector plates 114a, 114b and received into plates 44. The connection to drill stem 40a may be further strengthened and made rigid by clamping halves 112a, 112b together with fasteners through flanges 118a, 118b. Similarly, casing interface 120 is shown with halves 121a, 121b separated. In this configuration, halves 121a, 121b may be placed about drill stem 40a (between drill stem attachment 110 and cone 62 of casing 60) and fastened together using fasteners through flanges 128a, 128b.

(48) Thus, in an embodiment, drill stem attachment half 112a is rigidly fixed to casing interface half 121a and drill stem attachment half 112b is rigidly fixed to casing interface half 121b. For example, flange 116a may be welded to inner ring section 126 and outer ring section 124. Thus, when the two such joined sections 112a, 121a and 112b, 121b are placed about drill stem 40a, fastened to drill stem 40a, and fastened together using flanges 118a, 118b, 128a, 128b, rotation of drill stem 40a will be imparted to both drill stem attachment 110 and casing interface 120. Thus, when posts 122 are within cone slots 64, rotation of drill stem 40a will force casing 60 to rotate.

(49) FIG. 12 is a side view of casing driver 101 including perforated spacer 130. In FIG. 12, perforated spacer 130 includes posts 132, slots 134, and perforations 136. Slots 134 are shown to be configured similarly to cone slots 64, which allows spacer 130 to be used without modifying casing interface 120. Similarly, posts 132 are shown to be configured similarly to posts 122, which allows spacer 130 to be used without modifying cone 62. In the embodiment, perforated space 130 includes an upper section 138a with a diameter configured to receive casing interface 120 and a lower section 138b with a diameter configured to be received by cone 62. In embodiments, the length of sections 138a, 138b and the number and size of perforations 136 may be changed

arbitrarily as desired by the operator to accommodate spoils of different volume flow rates, dimensions, and consistencies. For drilling applications that require exhausting rather than displacing drill cuttings, incorporating the perforated spacer **130** allows a drill operator to exhaust drill cuttings while drilling.

(50) FIG. **13A** is an isometric view of the upper section of cone **62**. FIG. **13B** is an isometric view of the lower section of cone **62**. Although not shown for clarity, cone **62** is fixed to casing **60**. The means for fixing cone **62** to casing **60** may include welding, machine threading, and bolting. In embodiments, a casing driver **100**, e.g., as described above with reference to casing driver **101** (FIG. **8**-FIG. **12**) or below with reference to casing driver **102** (FIG. **14**-FIG. **21**), may interface with cone **62** of casing **60** by engaging slots **64** with, e.g., posts **122** of casing interface **120** or posts **132** of perforated spacer **130**. As shown, slots **64** allow posts **122**, **132** to enter through a top section and descend to a wider lower section. Thus descended, posts **122**, **132** may rotate slightly within slots **64** without imparting that rotation to cone **62**. When slightly rotated, posts **122**, **132** may engage a side lobe of slot **64**, which prevents withdrawing posts **122**, **132** upward from slot **64**. Thus, when casing interface **120** is connected to drill stem attachment **110**, drill stem **40a** may be used to pull cone **62** and casing **60** upward and, e.g., out of bore **80**.

(51) FIG. **14** is an upper isometric view of a second embodiment **102** of casing driver **100**. In FIG. **14**, casing driver **102** is adapted to a different drill stem **40**, i.e., a square drill stem **40b**.

(52) In FIG. **14**, casing driver **102** includes a drill stem attachment **150** and a casing interface **120**. Drill stem attachment **150** is connected to drill stem **40b** using a pin **146** (FIG. **15C**) that passes through drill stem attachment **150** and into one of slots **144a** . . . **144e** of drill stem **40b**. Thus, drill stem attachment **150** is connected to drill stem **40b** so that both rotation and axial movement of drill stem **40b** is imparted to drill stem attachment **150** (FIG. **16**). As with casing driver **101**, drill stem **40b** passes through casing driver **102** and into cone **62** and casing **60**.

(53) In the embodiment, drill stem attachment **150** is fixed to casing interface **120**, for example, by welding. Thus, linear force is transmitted from stem **40b** through drill stem attachment **150** to casing **60**, causing casing **60** to travel with drill stem **40b** into the bore. Likewise, rotation of drill stem **40b** is transmitted from drill stem attachment **150** to casing interface **120** and into casing **60**.

(54) In other embodiments, drill stem attachment **150** abuts casing interface **120** but is not otherwise connected to casing interface **120**. Thus, linear force is transmitted from stem **40b** through drill stem attachment **150** to casing **60**, causing casing **60** to travel with drill stem **40b** into the bore. However, in the embodiment, rotation of drill stem **40b** is not transmitted from drill stem attachment **150** to casing interface **120** because drill stem attachment **150** spins atop casing interface **120**.

(55) As with casing driver **101**, with casing driver **102**, casing interface **120** includes posts **122** that extend radially and are received by slots **64**, e.g., T-slots, within cone **62**. Thus, in the embodiment, casing interface **120** partially enters cone **62** to the depth that posts **122** descend into slots **64**.

(56) FIG. **15A** is a side view of drill stem **40b**. FIG. **15B** is a top view of the section indicated in FIG. **15A**. In FIG. **15A** and FIG. **15B**, drill stem **40b** is shown to be provided with slots **144a** . . . **144b** that are configured to receive pin **146** (FIG. **15C**). FIG. **15A** illustrates a single section of drill stem **40b** with a male upper connecting section and a female lower connecting section. Multiple such sections may be combined, one atop the other, depending on the desired depth of bore **80**. In addition, while FIG. **15A** illustrates two sets of slots **144a**, **144b** through drill stem **40b**, in embodiments, an arbitrary number of slots **144** may be provided in drill stem **40b** at arbitrary locations according to the desires of the operator. Providing two or more slots **144** spaced along the long axis of a drill stem **40b** enables the use of varying lengths of casing **60** without a drill operator needing to reconfigure or disassemble the drill string.

(57) FIG. **15B** further illustrates that a grout conduit **142** within drill stem **40b** is flexible enough to be moved aside within drill stem **40b** so that pin **146** may pass from one slot, e.g., slot **144b** through the center of drill stem **40b** and through the corresponding other slot, e.g., slot **144b**. The

direction of drill conduit **142** is flexed may alternate to prevent unbalanced rotation of the drill stem **40b** during drilling operations.

(58) FIG. **16** is an isometric view of casing driver **102** separated from cone **62**. In FIG. **16**, drill stem attachment **150** is fixed to casing interface **120**. The means for fixing drill stem attachment **150** to casing interface **120** may include welding, a threaded connection or a bolted connection. Drill stem attachment **150** includes identical attachment halves **152a**, **152b**. Each half **152a**, **152b** includes an upper flange **156a** and a lower flange **156b** connected by two connector plates **155** and by connector flanges **158a**, **158b**. Halves **152a**, **152b** are joined together, e.g., using fasteners through holes **159** in flanges **158a**, **158b**. Each connector plate **155** includes a pin slot **154**. To fix drill stem attachment **150** at a certain longitudinal location of drill stem **40b**, pin **146** is passed through a first pin slot **154**, through the two pin slots **144** at the desired location on drill stem **40b**, and through a second pin slot **154** of drill stem attachment **150**. Thus, drill stem attachment **150** is fixed longitudinally on drill stem **40b** and axially about drill stem attachment **150**. In addition, upper flange **156a** and lower flange **156b** are configured to receive square drill stem **40b** so that rotation of drill stem **40b** is transferred to drill stem attachment **150** by the corners of drill stem **40b** contacting and transferring torque to flanges **156a**, **156b**.

(59) FIG. **17** is an upper isometric view of casing driver **102** partially disassembled. In FIG. **17**, drill stem attachment **150** is shown with halves **152a**, **152b** separated. Similarly, casing interface **120** is shown with halves **121a**, **121b** separated. Attachment halves **152a**, **152b** are shown to be atop and fixed to interface halves **121a**, **121b**. The means to connect each respective attachment half **152a**, **152b** and to the respective interface half **121a**, **121b** may include welding, a threaded connection or a bolted connection. In other embodiments, these halves are not fixed together and rotation of drill stem attachment **150** is not imparted to casing interface **120**, since drill stem attachment **150** may spin atop casing interface **120**.

(60) In the configuration shown in FIG. **17**, halves **152a**, **152b** may be placed about drill stem **40b** and a pin **146** passed through a pair of pin slots **154** of drill stem attachment **150** and a pair of pin slots **144** of drill stem **40b**. The connection to drill stem **40b** may be further strengthened and made rigid by clamping halves **152a**, **152b** together with fasteners through flanges **158a**, **158b**. Similarly, casing interface **120** may be connected about drill stem **40b** in the manner discussed previously with regard to drill stem **40a**.

(61) Thus, when drilling a bore, drill stem **40b** may be used to drive flanges **156b** of drive drill stem attachment **150** against inner ring sections **126** of casing interface **120**. Because, in the embodiment, flanges **156b** are connected or fixed, e.g., by welding, to inner ring sections **126**, drill stem attachment **150** and casing interface **120** rotate together about the drill axis and transfer torsional, upward, and downward force into cone **62** and casing **60**.

(62) As a result, drill stem attachment half **152a** is rigidly fixed to casing interface half **121a** and drill stem attachment half **152b** is rigidly fixed to casing interface half **121b**. For example, flange **156a** may be welded to inner ring section **126** and outer ring section **124**. Thus, when the two such joined sections **152a**, **121a** and **152b**, **121b** are placed about drill stem **40b**, pinned to drill stem **40b**, and fasted together using flanges **158a**, **158b**, **128a**, **128b**, rotation of drill stem **40b** will be imparted to both drill stem attachment **150** and casing interface **120**. Thus, when posts **122** are within cone slots **64**, rotation of drill stem **40b** will force casing **60** to rotate.

(63) FIG. **18** is an upper isometric view of aspects of casing driver **102**. In FIG. **18**, a spacer **160** is positioned between casing interface **120** and cone **62**. In operation, the spacer **160** is a removable piece that enables a drill operator to lower the casing driver **102** onto the spacer **160** to place it into a resting position so that support personnel can either unbolt and remove the casing driver **102** or attach it to the drill stem **40b**. When preparing to attach or remove the casing driver **102**, posts **122** rest atop spacer **160**. In this position, posts **122** do not engage cone slots **64** making the casing driver **102** easier and safer for support personnel to install casing driver **102** onto drill stem **40b** or remove casing driver **102** from drill stem **40b**.

(64) FIG. 19 is a side view of the casing driver 102 with optional spacer 160. In FIG. 19, casing interface 120 is shown to include an additional flange 129 atop inner ring sections 126. Flange 129 may be split like rings 124, 126 to provide two identical halves. Flange 129 may be used to provide additional contact area for flanges 156a, 156b. Also, like flanges 156a, 156b, flange 129 is also configured to receive square drill stem 40b so that rotation of drill stem 40b is transferred to casing interface 120. As shown in FIG. 19, the taper of posts 122 may help center casing interface 120 and drill stem 40b within spacer 160, and, as a result, within cone 62. In turn, posts 162 and 163 are configured to fit into slots 64 so that no rotation is allowed between spacer 160 and cone 62. When spacer 160 is installed around drill stem 40b the casing driver 102 can rest on top of spacer 160 such that posts 122 do not engage slots 64. This helps support personnel attach and remove casing driver 102 from drill stem 40b. During drilling operations, spacer 160 is removed.

(65) FIG. 20 is an upper isometric view spacer 160 disassembled. In FIG. 20, each half ring 164 of spacer 160 is shown to include posts 162 which are configured to fit into slots 64 and prevent spacer 160 from rotating relative to cone 62. In addition, each half ring 164 includes split connector plate/spacer posts 163, which, when half rings 164 are brought together, both posts 163 fit into a single slot 64 and further prevent spacer 160 from rotating relative to cone 62. Thus, in some embodiments, half rings 164 do not need to be bolted together. Spacer 160 includes a taper 166 on the upper edge of half rings 164. Taper 166 facilitates the centering of casing interface 120 into spacer 160. Spacer 160 may also include one or more centralizing bars 165 fixed to either side of post 162 and on at least one side of post 163 with an offset between the centralizing bar and the outer wall surface of half ring 14. These centralizing bars centralize spacer 160 in slot 64 and prevent posts 162 and 163 of spacer 160 from sliding out of the vertical leg of slot 64. In some embodiments posts 163 each include openings configured to bolt each half ring 164 together or to receive rigging and lifting equipment such as shackles, chains, slings or straps.

(66) FIG. 21 is a side view of casing driver 102 including perforated spacer 130, described earlier with reference to FIG. 12. The use of perforated space 130 may be used with casing driver 102 in the manner described with reference to casing drivers 100 and 101.

(67) In the description above, connector plates 44, connector plates 114, and fasteners were described for attaching casing driver 101 to the drill stem and slots 144, connector plates 154, and pin 146 were described for attaching casing driver 102 to the drill stem. However, in embodiments, these apparatuses for attaching a casing drive to a drill stem may each be used for any embodiment, e.g., casing driver 101 and drill stem 40a may be configured to use pin 146 and casing drive 102 and drill stem 40b may be configured to use fasteners and connector plates. Similarly, elements described with regard to one embodiment may be adapted for use with a different embodiment. For example, spacer 160 may be used with casing driver 101 as well as casing drive 102.

(68) To facilitate an understanding of the subject matter described, many aspects are described in terms of sequences of actions. The description herein of any sequence of actions is not intended to imply that the specific order described for performing that sequence must be followed. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly.

(69) While one or more implementations have been described by way of example and in terms of the specific embodiments, it is to be understood that one or more implementations are not limited to the disclosed embodiments. To the contrary, it is intended to cover various modifications and similar arrangements as would be apparent to those skilled in the art. For example, one skilled in the art will recognize that these embodiments can be practiced without one or more of the specific details, or with other components, systems, etc. And, in other instances, there may be structures or operations not shown, or not described in detail, to avoid obscuring aspects of the described embodiments. Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

(70) The previous description is provided to enable any person skilled in the art to practice the

various aspects described herein. In the embodiments, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the subject disclosure.

(71) A phrase such as an “aspect” does not imply that such aspect is essential to the subject technology or that such aspect applies to all configurations of the subject technology. A disclosure relating to an aspect may apply to all configurations, or one or more configurations. A phrase such as an aspect may refer to one or more aspects and vice versa. A phrase such as a “configuration” does not imply that such configuration is essential to the subject technology or that such configuration applies to all configurations of the subject technology. A disclosure relating to a configuration may apply to all configurations, or one or more configurations. A phrase such as a configuration may refer to one or more configurations and vice versa.

(72) All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims.

Claims

1. An apparatus comprising: a drive head having an attachment section configured to mount to a drill stem, the attachment section including a first attachment plate having one or more perforations, each of the one or more perforations configured to accommodate a fastener for connecting the attachment section to the drill stem; a casing interface section connected to the attachment section such that rotation of the attachment section about an axial direction is imparted to the casing interface; and a casing, wherein when the drill stem is connected to the attachment section and the casing interface is positioned between the attachment section and the casing, a first force applied to the drill stem in the axial direction toward the casing is transmitted through the attachment section to the casing interface and to the casing.
2. The apparatus of claim 1, wherein the attachment section includes first and second parts that separate to accommodate the drill stem between the first and second parts.
3. The apparatus of claim 2, wherein joining the first and second parts about the drill stem creates a clamping force that hinders relative motion in the axial direction between the drill stem and the attachment section.
4. The apparatus of claim 1, further including the drill stem and the drill stem including a plurality of second attachment plates configured to receive the fastener, the plurality of second attachment plates distributed longitudinally along the drill stem.
5. The apparatus of claim 1, wherein the attachment section includes a first flange and the casing interface section includes a second flange, and, when the casing interface is between the attachment section and the casing, the first flange and the second flange abut each other.
6. The apparatus of claim 5, wherein the casing interface section being connected to the attachment section such that rotation of the attachment section about the axial direction is imparted to the casing interface includes the first flange and the second flange being connected to each other.
7. An apparatus comprising: a drive head having an attachment section configured to mount to a drill stem; and a casing interface section connected to the attachment section, wherein the

attachment section includes a attachment plate having one or more perforations, each of the one or more perforations configured to accommodate a pin, which when passed through any of the one or more perforations into the drill stem, hinders relative motion between the drill stem and the attachment section in an axial direction such that rotation of the attachment section about the axial direction is imparted to the casing interface section; a casing, wherein when the drill stem is connected to the attachment section and the casing interface section is positioned between the attachment section and the casing, a first force applied to the drill stem in the axial direction toward the casing is transmitted through the attachment section to the casing interface section and to the casing.

8. The apparatus of claim 7, further including the drill stem and the drill stem including a plurality of perforations distributed longitudinally along the drill stem and configured to receive the pin.

9. An apparatus comprising: a drive head having an attachment section configured to mount to a drill stem; a casing interface section connected to the attachment section such that rotation of the attachment section about an axial direction is imparted to the casing interface section, the casing interface section including a first plurality of posts extending radially relative to the axial direction; and a casing, wherein when the drill stem is connected to the attachment section and the casing interface section is positioned between the attachment section and the casing, a first force applied to the drill stem in the axial direction toward the casing is transmitted through the attachment section to the casing interface section and to the casing.

10. The apparatus of claim 9, wherein the casing includes a plurality of slots configured to receive the first plurality of posts.

11. The apparatus of claim 9, wherein the casing interface section includes a flange comprised of separable first and second flange sections, a first subset of the first plurality of posts connects a first inner cylinder section to a first outer cylinder section, and a second subset of the first plurality of posts connects the second inner cylinder section to the second outer cylinder section.

12. An apparatus comprising: a drive head having an attachment section configured to mount to a drill stem; and a casing interface section connected to the attachment section such that rotation of the attachment section about an axial direction is imparted to the casing interface section, the casing interface section includes an inner cylinder comprised of separable first and second inner cylinder sections and an outer cylinder comprised of separable first and second outer cylinder sections; and a casing, wherein when the drill stem is connected to the attachment section and the casing interface section is positioned between the attachment section and the casing, a first force applied to the drill stem in the axial direction toward the casing is transmitted through the attachment section to the casing interface section and to the casing.

13. The apparatus of claim 12, wherein the casing interface includes a flange comprised of separable first and second flange sections, the first inner cylinder section and the first outer cylinder section are connected together by the first flange section, and the second inner cylinder section and the second outer cylinder section are connected together by the second flange section.

14. An apparatus comprising: a drive head having an attachment section configured to mount to a drill stem, the drive head further including a spacer section having a plurality of posts configured to be received by slots in a casing; and a casing interface section connected to the attachment section such that rotation of the attachment section about an axial direction is imparted to the casing interface section, wherein when the drill stem is connected to the attachment section, the casing interface section is between the attachment section and the spacer section, and the spacer section is between the casing interface section and the casing, a first force applied to the drill stem in the axial direction is transmitted to the casing through the attachment section, the casing interface section and the spacer section.

15. The apparatus of claim 14, wherein the spacer section includes a plurality of perforations.

16. A method comprising: connecting a rotary drive of a drilling rig to a drill stem at a first longitudinal location on the drill stem, the drill stem including a drill bit at a distal end; connecting

a drive head to the drill stem at a second longitudinal location on the drill stem separated from the first longitudinal location placing a casing about the drill stem, the casing extending from the drive head toward the drill bit; drilling a first length of a bore with the casing and drill stem moving into the bore and without relative axial motion between the casing and drill stem; disconnecting the drive head from the drill stem; and drilling a second length of the bore with the drill stem moving axially within the casing and the casing remaining within the first length of the bore.

17. The method of claim 16, further comprising, after drilling the first length of the bore and before disconnecting the drive head from the drill stem: connecting the rotary drive at a third longitudinal location on the drill stem separated from the first and second longitudinal locations.

18. The method of claim 16, further comprising; pumping grout into the bore; withdrawing the drill stem from the second length of the bore with the drill stem moving axially within the casing and the grout being pumped into the bore at a first volume flow rate that causes the grout to fill the bore to match a first withdrawal rate of the drill stem; reconnecting the drive head to the drill stem; engaging the drive head and the casing; and withdrawing the drill stem and casing from the first length of the bore while further pumping the grout into the bore, without relative axial motion between the casing and drill stem, and with the grout being pumped into the bore at a second volume flow rate that causes the grout to fill the bore to match a second withdrawal rate of the drill stem.
